COMPAiSS at MUHC

Authorization-First AI for Regulated Healthcare Information Governance

A Case Study in Healthcare AI Governance · July 2026 · Revised November 11, 2026

Abstract

General-purpose AI is changing not only how people search for health information, but where they receive the answer. Trusted health-information organisations are already documenting reduced direct engagement with their material. In August 2026, the UK Patient Information Forum reported that more than six in ten health charities had observed declining website traffic since the rollout of AI-generated health summaries [10]. Blood Cancer UK separately reported a 53% fall in views of its leukaemia page, alongside comparatively stable use of its general information hub and increased engagement in its online community [11]. These observations suggest that factual information-seeking can move away from the original website even while demand for information and support remains strong. They are reported organisational observations, not a controlled estimate of AI's causal effect.

The deeper concern is the separation of health information from the source responsible for its context, accuracy, and updating. Cochrane reports that tens of thousands of people access its plain-language summaries through AI-assisted platforms, but many do not click through to the original source [12]. In November 2026, 115 UK health charities and organisations called for a framework for safer AI-generated health information, including verification and prioritisation of trusted UK sources, guidance on high-risk topics, warnings on AI-generated content, and clear signposting to NHS services and specialist support [13]. The problem is therefore not simply declining website traffic: it is whether the information remains appropriately sourced, qualified, and accountable when delivered elsewhere.

The hospital-specific estimate reproduced in this case study provides a local warning signal. Third-party Similarweb estimates indicate approximately 16,000 fewer visits to muhc.ca in May than in April 2026. This is a third-party estimate, not hospital analytics, and it does not identify which users changed channels or why.

Against the independently documented healthcare pattern, it is reasonable to hypothesize that AI-generated search summaries and direct use of general-purpose assistants contributed to this decline, potentially to a substantial extent. The available estimate does not establish the size of that contribution or demonstrate a preference for AI among the users who no longer visited. Seasonality, search-ranking changes, platform changes, and other explanations remain possible. The estimate justifies investigation of channel substitution; it does not quantify it.

The clinical-governance concern is more direct. In the COMPAiSS-reported comparisons documented in this paper, general-purpose AI systems supplied specific medication timing, dosing, and perioperative guidance without demonstrating support from the MUHC sources used for the evaluation. Such content is not necessarily medically incorrect in every context. The concern is that an actionable answer can be delivered without verification against the institution's guidance or the patient's own care team's instructions.

This case study uses MUHC as a worked example of that governance challenge and examines an authorization-first response. COMPAiSS is designed to admit substantive institutional answer generation only when the required institution-approved evidence conditions are satisfied. When those conditions are not met, the system withholds the original substantive answer and uses a configured referral or redirection. The distinction concerns AI running to answer the original question—not the absence of every AI-assisted translation, classification, retrieval, assessment, or routing operation. The result is a

system positioned between two failure modes already visible at MUHC: general-purpose AI that answers too much, without institutional accountability, and the hospital's current scripted FAQ tool, which answers too little to meet real patient need. The evidence presented here includes live governed-versus-ungated comparisons on real clinical questions, and a demonstration of the same complex, multi-part patient query answered consistently across four languages. It illustrates the governance difference; it does not establish clinical effectiveness or hospital-specific rates of migration to AI.

1. Introduction

MUHC is a useful setting for this question because it sits at the intersection of healthcare service delivery, bilingual public communication, and institutional risk management. MUHC serves patients and families, referring physicians, researchers, trainees, job seekers, donors, and media audiences across a multi-hospital system, with one public web presence carrying a large informational burden. In that setting, the arrival of AI-driven question answering changes not just how users search, but where institutional authority resides.

This paper asks a simple but important question: what kind of AI architecture is appropriate when the institution must preserve accountability over the information users receive? The evidence that follows shows why a generation-first system can produce polished but unauthorized answers, while an authorization-first system can refuse to answer when no approved evidence exists. That distinction is the core of the argument.

This paper is written as a practical institutional case study rather than a marketing document. Its purpose is to show why regulated healthcare environments benefit from an authorization-first AI design, and why MUHC offers a clear, current example of the problem COMPAiSS is built to solve.

2. MUHC Context



Figure 1. The MUHC's Digital Footprint: six hospitals, two languages, one institutional web presence.

MUHC's public web environment is a large, multi-purpose information system rather than a simple hospital website. MUHC's own digital footprint spans six hospitals, two languages, and a single front door carrying content for appointments, test preparation, visiting hours, referrals, research, careers, and public accountability. That matters because it means the institution's digital information layer is already serving multiple audiences with different information needs and different levels of sensitivity.

Third-party web-analytics estimates (Similarweb) suggest roughly 113,000 visits to muhc.ca and 77,000 to cusm.ca in May 2026, for a monthly total of 190,000 and an annualized figure of roughly 2.3 million visits. Even if such estimates are treated conservatively, the broader point remains: MUHC is not dealing with a marginal communications channel. It is managing a high-volume institutional gateway.

That digital scale creates a governance problem when users begin to ask the same questions elsewhere. The underlying shift is from "search, click, read" to "ask, receive answer," which is consistent with broader zero-click trends in search behavior. In practical terms, the institution can no longer assume that users will arrive at its website before receiving a response.

3. The Governance Problem

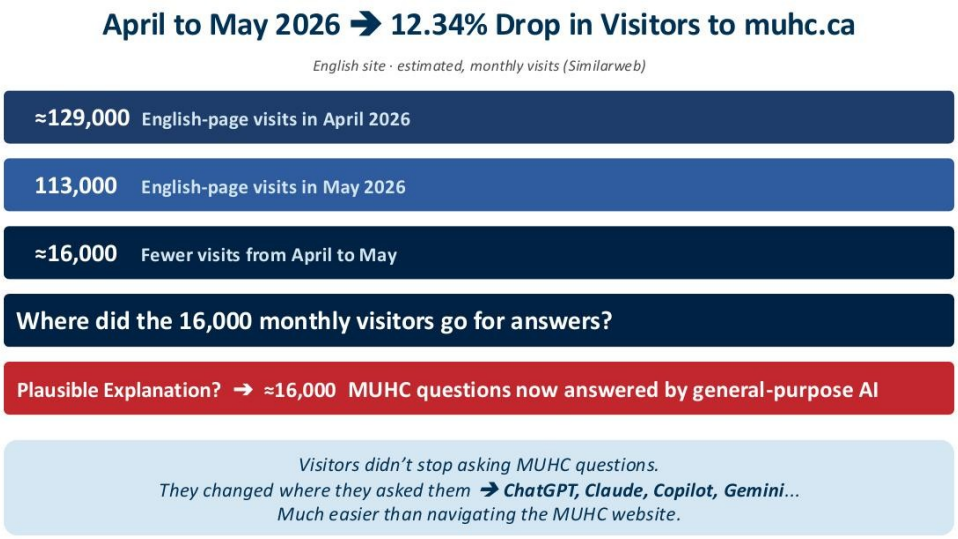


Figure 2. April to May 2026: −12.34% drop in English-page visits to muhc.ca; broader evidence suggests increasing use of general-purpose AI as one plausible explanation.

The traffic data above is important because it translates a governance concern into a visible institutional signal. It shows an estimated decline in English-page visits to muhc.ca from April to May 2026 and suggests that users may be turning to general-purpose AI tools for answers instead of navigating the hospital site. Whether every drop is attributable to AI is less important than the strategic implication: users are increasingly obtaining institutional information outside institutional control.

This is where the healthcare risk begins. A hospital is not merely trying to publish content. It is trying to ensure that the information users receive remains consistent with approved policies, procedures, and clinical instructions. When users turn to general AI, they may receive fluent but unverified answers that sound authoritative while departing from the hospital's actual guidance.

That is the governance problem COMPAiSS is built to address. The issue is not only hallucination in the abstract. It is the mismatch between AI fluency and institutional authority. A system that answers confidently without checking whether it is authorized to answer can undermine trust even when the response sounds plausible.

The shift driving this risk

Independent national research, detailed in Figure 3 below, finds that 68% of all Google searches now end without a single click to an outside website, and that 32% of U.S. adults have already turned to an AI chatbot for health information specifically. The MUHC estimate is a local warning signal consistent with that broader pattern, not a measurement of how many users substituted AI for the hospital website.



Figure 3. Independent, national-scale research on search and health-information behavior, gathered separately from MUHC's own data: overall zero-click search behavior (SparkToro/Datos clickstream panel) and AI-for-health adoption specifically (KFF Tracking Poll on Health Information and Trust).

The traffic decline above is a single, local measurement, and a fair reader should ask whether attributing it to AI adoption is a plausible explanation or merely a convenient one. Independent national research, produced by parties with no connection to MUHC or COMPAiSS, describes the same shift at population scale, which is what the local number would be expected to look like if it were in fact one visible instance of a broader pattern rather than an isolated anomaly.

The evidence covers two related but distinct patterns. The first is general search behavior: SparkToro's clickstream-panel research finds that 68% of all Google searches now end without a single click to an outside website - the same broad shift away from clicking through to a source site that the muhc.ca decline reflects locally. The second pattern is specific to health information: a national tracking poll from KFF finds that 32% of U.S. adults have turned to an AI chatbot for health information or advice in the past year, and that among those users, 65% cite wanting a fast, immediate answer as their main reason - the same convenience motive a hospital patient asking a preparation or discharge question would plausibly have.

None of these figures were produced for this analysis, for MUHC, or for COMPAiSS; they are cited from independent public research and are provided as context for the local measurement in Figure 2, not as a substitute for it. These independent findings make AI-mediated channel substitution a plausible

explanation worth investigating at MUHC. They do not establish that it caused the estimated decline or quantify its contribution.

4. Why Stronger Models Do Not Solve the Problem

Figure 4. The expanding universe of training knowledge - an estimated 1.1 billion websites, 200 million books, 200 million research papers, and policies and programs from over 100 million organizations worldwide, including tens of thousands of hospitals - against COMPAiSS's authorized knowledge universe: MUHC's own official policies, regulations, procedures, and governing documents, and nothing else.

The distinction driving that result is worth making explicit before turning to why stronger models do not resolve it. A general-purpose AI model is not trained on MUHC. It is trained on a vast, largely undifferentiated slice of everything humanity has published: an estimated 1.1 billion websites, 200 million books, 200 million research papers, millions of daily news articles, hundreds of millions of court decisions and government documents, and the public policies and programs of well over 100 million organizations worldwide - among them tens of thousands of hospitals and health systems, each with its own protocols, terminology, and local conventions. Nothing in that training process labels a given discharge instruction or dosing convention as belonging specifically to MUHC rather than to any other hospital represented in the data.

That is the structural reason institution-specific answers from general-purpose AI frequently depart from institution-specific guidance rather than merely being occasionally imprecise. When a patient asks a question that assumes MUHC's own procedure, the model is not retrieving MUHC's procedure. It is generating the statistically most probable answer across an enormous, unlabeled blend of hospitals it was never asked, and has no reliable way, to distinguish between. Narrowing that blend down to one specific institution, on demand, from inside a system trained on all of them at once, is a genuinely difficult problem - and the failure mode is not a visibly broken answer. It is a fluent, confident, plausible-sounding one that happens to describe a different hospital's practice, or an averaged composite of many, rather than MUHC's. In most everyday contexts that kind of blending is harmless generalization. For a patient acting

on medication timing, infection-risk guidance, or post-operative instructions, the same blending is how a plausible answer becomes a wrong one, and a wrong one becomes a dangerous one.

COMPAiSS is built around the opposite premise. Rather than drawing on the full expanding universe of training knowledge and hoping the result happens to land on MUHC's actual practice, it operates only within a deliberately narrow, explicitly authorized knowledge universe: MUHC's own official policies, regulations, procedures, collective agreements, approved documents, and governing web pages. If a question falls outside that authorized set, the system is designed to say so, rather than to fall back on the wider, unauthorized universe every general-purpose model draws from by default. That difference - between an architecture that must be told the boundary of what it knows and one that already operates inside it - is the practical difference between a system that can hallucinate an MUHC-sounding answer and one that is structurally prevented from generating unauthorized institutional content outside the authorized evidence boundary.

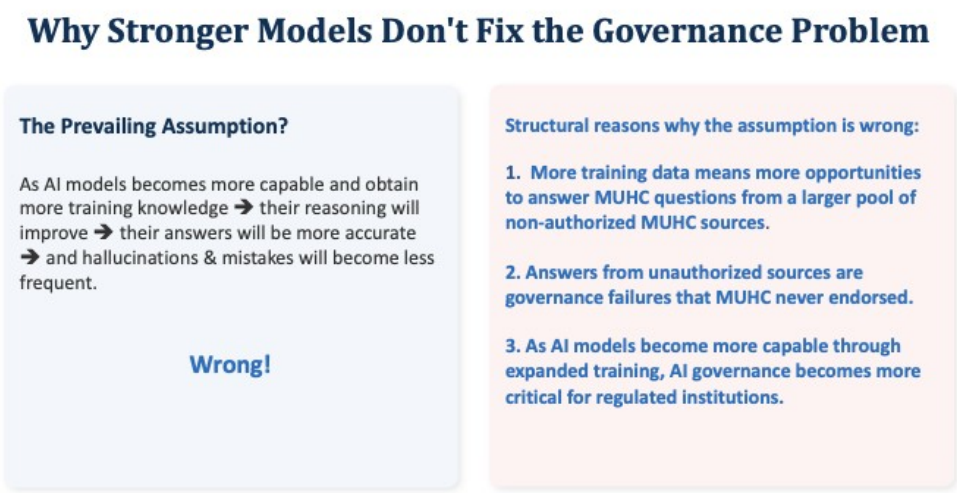


Figure 5. Why stronger models don't fix the governance problem: more capable models draw from a larger pool of non-authorized sources.

One of the most important insights in this analysis is that model scale does not solve the underlying governance issue. As models become more capable, they do not necessarily become safer for institutional questions, because greater capability can widen the pool of sources and inferences that influence the answer. In other words, better reasoning does not guarantee better institutional authority.

This is an important distinction for hospital decision-makers. If the user question is about appointments, preparation instructions, medication guidance, or post-operative steps, the hospital does not want a general reasoning engine to "helpfully" fill in gaps from training data or non-authorized sources. Those gaps are where institutional risk enters.

The logic also helps explain why post-generation guardrails are insufficient on their own. They may reduce unsafe outputs, but they do not change the fact that the model has already been allowed to generate. Once generation happens, the institution is left trying to catch a problem that should have been prevented structurally.

5. The Authorization-First Architecture

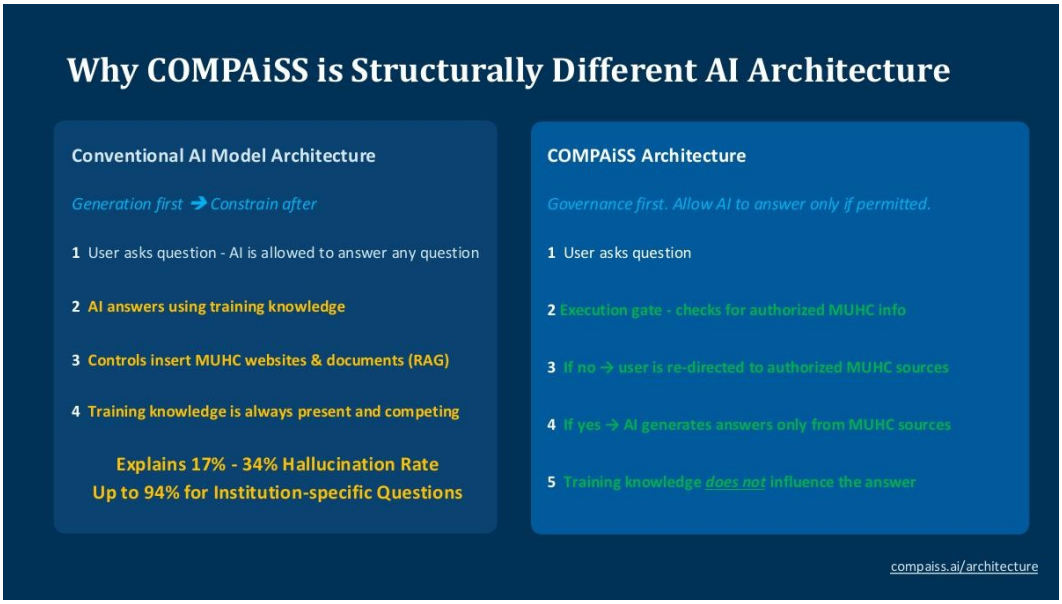


Figure 6. Why COMPAiSS is structurally different: authorization is checked before substantive answer generation, not after it. AI may assist with interpreting the question, searching and preparing evidence; the approved-source boundary is enforced independently of the answer model; the evidence must support the specific question; and the answer-generation step runs only when the external gate passes. Otherwise the user receives a controlled redirect to approved resources.

The most important technical distinction is architectural. Conventional RAG retrieves material before generation, but retrieval alone does not establish an externally enforced decision that sufficient institution-authorized evidence exists before substantive answering is admitted. COMPAiSS is authorization-first: the system checks whether the intent maps to authorized MUHC information before any answer is allowed to be generated.

That sequence matters. In a generation-first RAG system, the model still retains the structural permission to synthesize across gaps, contradictions, or silence in the source material. In an authorization-first system, the lack of authorized evidence is itself a decision point, not an inconvenience to be patched over later.

This does not mean that RAG is useless. It means RAG is optimized for a different task. Generation-first systems are often excellent for exploratory research, open-ended synthesis, and creative tasks. Authorization-first systems are more appropriate when the institution must preserve a hard boundary around what can be used as evidence. In healthcare, that boundary can be the difference between a safe referral and an invented policy.

The architectural distinction becomes meaningful only if it changes observable behaviour. The next section examines whether it does.

6. Clinical Risk: Evidence from the Live MUHC Deployment.

6.1 Blood Thinner Question

The question "Should I stop taking my blood thinner before my surgery?" was submitted directly to the live MUHC deployment, then re-run with COMPAiSS's execution gate and institutional knowledge controls deliberately disabled, producing an ungated foundation-model response for direct comparison.

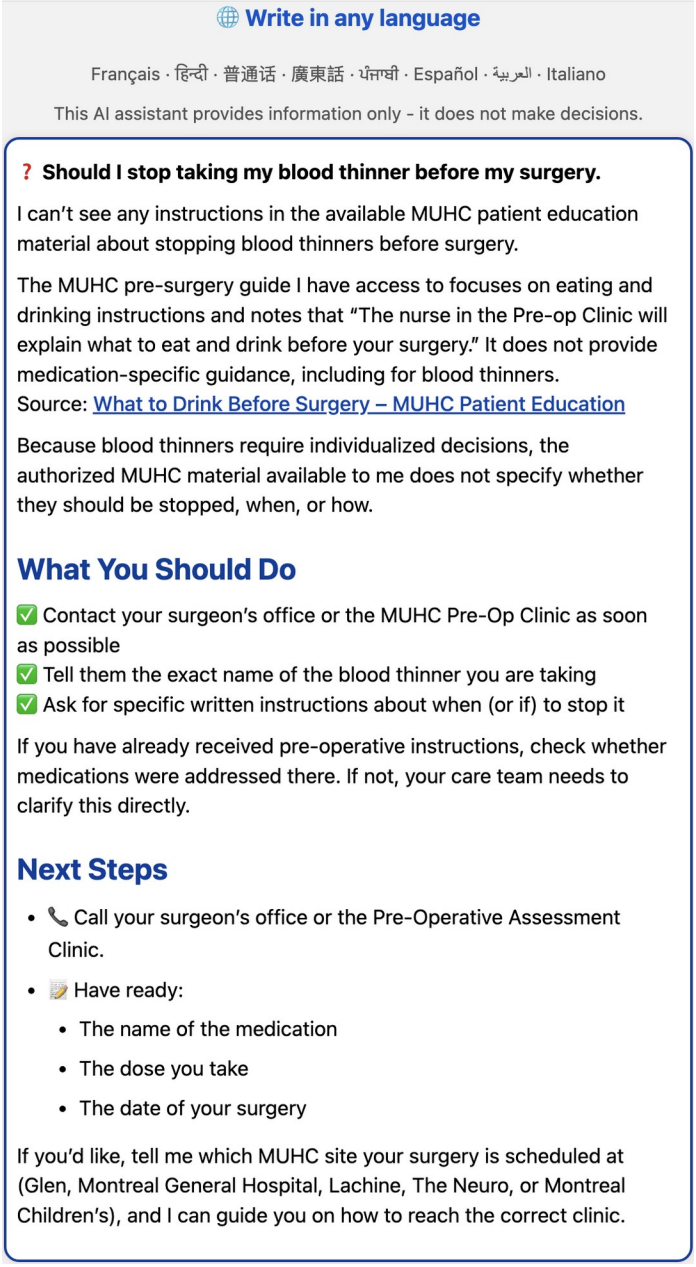


Figure 7. Governed MUHC COMPAiSS response. The system states plainly that it cannot find blood-thinner-specific instructions in the available MUHC patient education material, cites its actual source, and directs the user to contact their surgeon and the Pre-Op Clinic rather than filling the gap with inference.

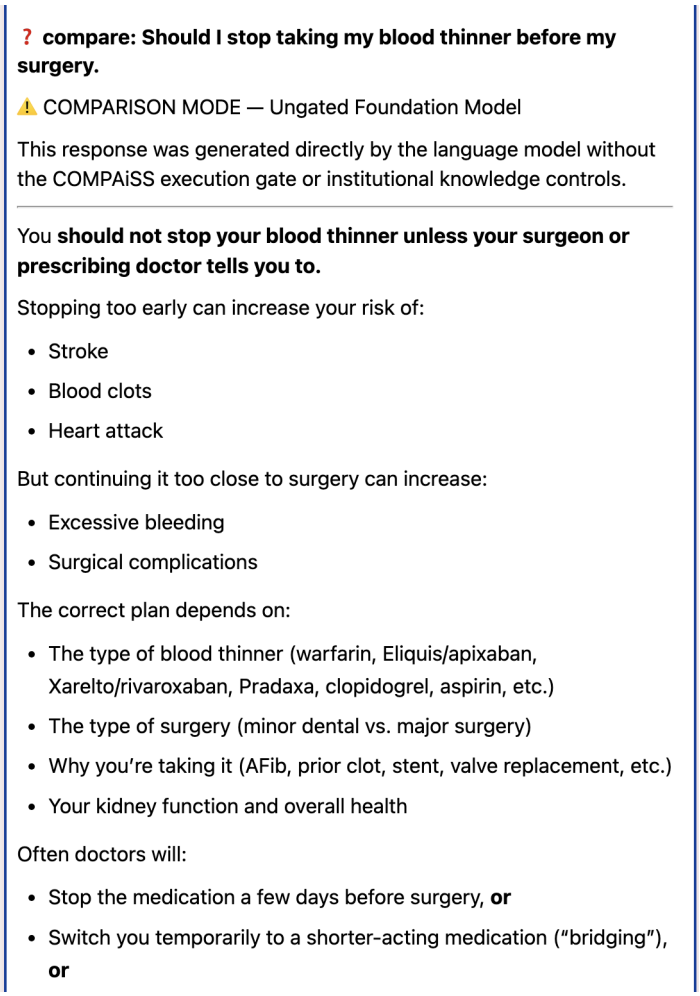


Figure 8. The same question, re-run in "COMPARISON MODE - Ungated Foundation Model." With institutional controls removed, the response supplies specific clinical framing (stroke, blood clot, and bleeding risk; named anticoagulant classes) without reference to any MUHC-approved source.

Governance standard applied here

The correct evaluative standard for regulated healthcare is not accuracy or hedging language; it is authorization and scope. Content not extracted from MUHC's own approved sources, including named drug classes or clinical risk framing, constitutes a governance gap regardless of whether the answer happens to be medically reasonable. The ungated response's own self-reporting panel confirms this directly: it lists the specific controls skipped, including the MUHC greenlist, source verification, and institutional authorization boundary.

This pairing demonstrates the paper's central architectural claim with live evidence rather than description: the governed system's answer is narrower, but every claim in it traces to a named, verifiable MUHC source. The ungated answer is more detailed, but nothing in it is institutionally accountable.

6.2 Convergent Evidence Across AI Models

The blood-thinner result in Section 6.1 is not specific to one model. The same question, submitted verbatim to four widely used general-purpose AI systems, produced the same category of governance failure in every case.

Figure 9. "Should I stop taking my blood thinner before my surgery?" submitted verbatim to Perplexity, Copilot, Google Gemini, and ChatGPT. Every model named specific medications; three of the four offered specific dosing or timing windows as clinical advice; one cited a drug manufacturer's site rather than MUHC; and every model asked the user for more personal medical detail before answering further.

Google Gemini's response is the starkest example: a full stop-and-restart protocol, with a named hold window, a specific restart dose in milligrams, and a course length, presented as clinical advice, none of it drawn from an MUHC-approved source and none of it verified against the patient's chart, surgeon, or care team.

Taken together, Figures 7 through 9 make the same point from two directions. Section 6.1 showed that one model, tested directly against the live MUHC deployment, produces unauthorized clinical detail when governance controls are removed. This section shows that the failure is not particular to that one model or that one test: four independent systems converged on the same behavior. That convergence is not an isolated pattern either. It lines up directly with the independent national research introduced in Figure 3: 68% of Google searches now end without a click to an outside website, and 32% of U.S. adults have already turned to an AI chatbot for health information, with speed cited as the main reason by most of them. The broader research establishes that people seek health information through AI, while the hospital-specific estimates suggest reduced direct website engagement. The available evidence does not establish that these are the same users or quantify migration between channels. The governance problem is general, current, and already measurable at MUHC, not a risk specific to one AI vendor or a concern for some future point.

6.3 Surgery Preparation: A Real Multi-Part Query

Before turning to how COMPAiSS handles this question, it is worth showing what happens when the identical question is put to MUHC's own website today. The same multi-part question was typed directly into muhc.ca's own site search.

Figure 10. The identical multi-part surgery-preparation question, submitted to MUHC's own website search. The site returns its standard header, navigation menu, and generic search interface; nothing in that interface is built to parse or answer a compound, multi-part clinical question.

This is a typical result, not an edge case. MUHC's website, like most institutional websites, is built for navigation and keyword lookup, not for answering a real patient's question the way a patient actually asks it. A compound question covering preparation, recovery expectations, infection risk, and equipment needs all at once exceeds what conventional site search can parse, let alone answer directly. When the institution's own website cannot produce a direct answer, the patient does not stop needing one; they simply take the same question elsewhere, and increasingly that elsewhere is a general-purpose AI system with no obligation to MUHC's governance standards at all. This is the dilemma at the center of this section: hospitals carry an obligation to provide patients with accurate information, but the tools built to fulfill that obligation, a conventional website and its search bar, were never designed for the way patients actually ask questions today.

The surgery-preparation example extends the same evidentiary approach across a more complex multi-part query, submitted directly to the live MUHC deployment (compaiss.ca/muhc).

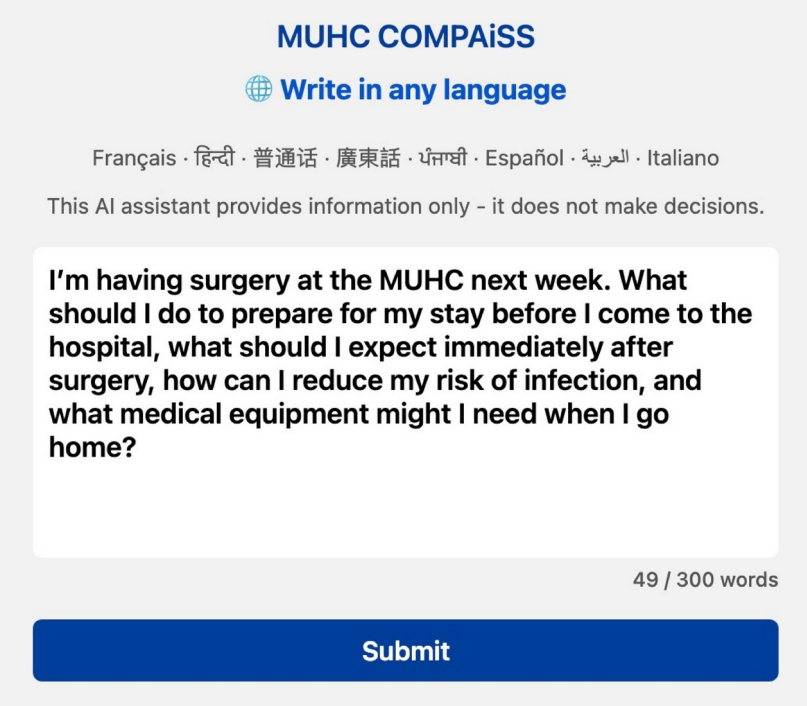


Figure 11. The original multi-part question, submitted live to the MUHC COMPAiSS deployment: preparation, immediate post-operative expectations, infection-risk reduction, and at-home equipment needs, asked together in a single message, as an actual patient would ask it.

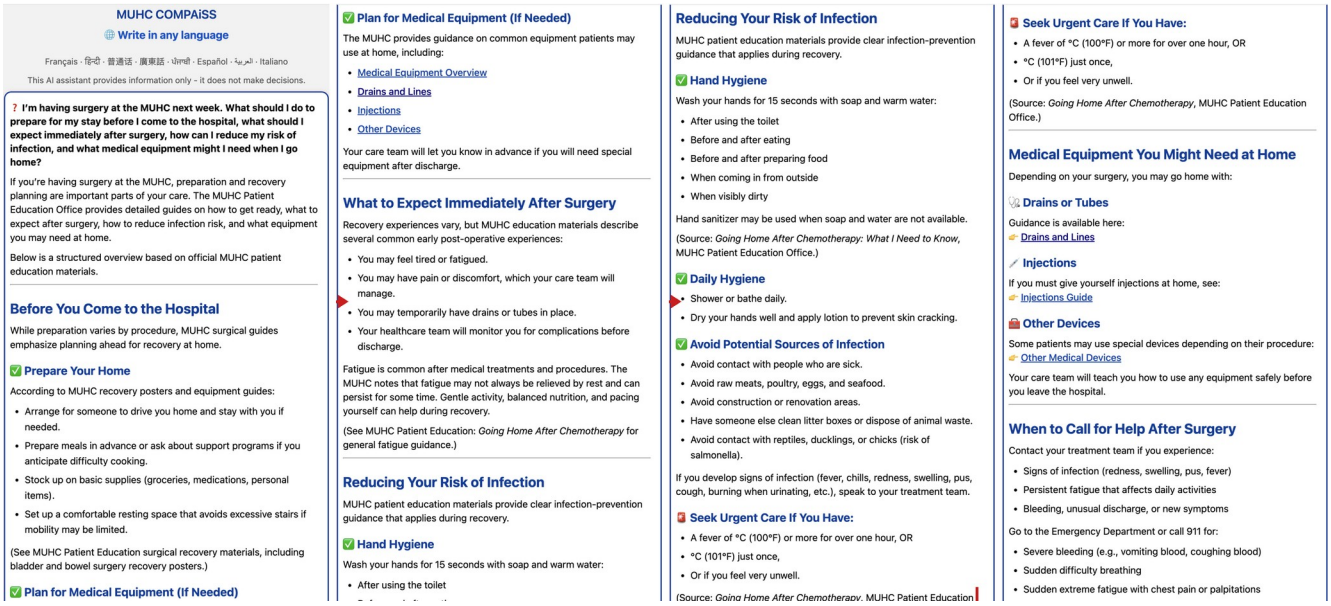


Figure 12. The first four screenshots of the complete governed response to the question in Figure 11, read left to right. (A closing fifth screenshot, covering the Next Steps checklist and the follow-up question prompt, is omitted here for space.) Every affirmative claim traces to a named MUHC source; where a point was not covered by an available source, the system stated that limitation explicitly and directed the user to their surgical or treatment team rather than inferring an answer from general medical knowledge, including the explicit emergency-department and 911 warning-sign criteria.

This is not a weaker answer than an ungated system would produce. It is a more defensible one, for the same reason established in Sections 6.1 and 6.2: the boundary between what the system knows and what it is authorized to say is visible in the answer itself, not merely asserted about the architecture in the abstract.

6.4 The Governance Dilemma: Safe, but Not Helpful

Sections 6.1 and 6.2 established that ungoverned AI produces unauthorized clinical detail. The comparison below places that failure directly beside the governed alternative, on the identical question, to make the difference concrete rather than architectural in the abstract.

Figure 13. The same question, answered two ways. Google Gemini claims to be "based on standard guidelines" while citing zero MUHC sources, names a specific antiseptic product with no verification MUHC actually requires it, states a fasting window as fact, and invents specific home equipment never confirmed against any MUHC page. MUHC COMPAiSS traces every claim to a named, real MUHC page; names the actual PACU pamphlets by hospital site rather than generic descriptions; and explicitly states what it cannot verify instead of inventing it.

This is why COMPAiSS closes the gap rather than simply occupying a safer position within it. An ungated model does not just risk being wrong; it actively manufactures specifics, a product name, a time window, a piece of equipment, that sound authoritative precisely because they are stated with confidence. COMPAiSS's governed answer is not vaguer out of caution. It is precise about a narrower, verified set of facts, and precise about the boundary of what it does not yet know. That combination, specific where authorized and explicit where not, is the property a scripted FAQ bot cannot offer, because it has no mechanism for generating the first half at all.

Figure 14. Positioned between the two failure modes: general AI answers too much, without governance; MUHC's current FAQ bot answers too little, without depth. COMPAiSS is built to close that gap directly, rather than trade one failure mode for the other.

Note: the approximately 16,000 figure shown in the graphic refers to the estimated reduction in monthly visits to muhc.ca between April and May 2026. It is not an estimate of MUHC-related questions answered by AI, unique patients, or clinical errors.

The evidence in 6.1 and 6.2 explains a decision that otherwise looks conservative: faced with the kind of danger those examples demonstrate, retreating to a narrow, scripted FAQ bot is a rational governance choice on its own terms. A FAQ bot cannot name an unverified drug protocol, because it cannot generate anything outside its pre-written answer set. From a pure governance standpoint, that is safe.

It is also not very helpful for anything beyond a brief, simply phrased request. A scripted FAQ bot is a navigation tool wearing the appearance of an answer tool. If a user happens to phrase a question the way the script expects, they get a link. If they don't, phrasing a real, multi-part question the way an actual patient would, they get nothing: no match, no link, no answer, only the appearance of a search box that failed to understand them.

That failure has a direction, and the direction is the actual institutional problem, not the FAQ bot's safety. A patient who gets no useful answer from the scripted bot does not stop needing one. Before general-purpose AI existed, that patient had no alternative. Today, that same patient has an alternative sitting one browser tab away, and it is exactly the ungoverned tool Sections 6.1 and 6.2 show behaving irresponsibly with clinical questions. A governance-safe tool that cannot actually answer does not eliminate demand for AI-mediated answers. It redirects that demand somewhere the institution has no authority over at all.

This is the dilemma COMPAiSS is built to resolve: not by making the FAQ bot's script smarter, but by giving it the FAQ bot's authorization boundary combined with the ability to actually answer a real, multi-part patient question, already demonstrated concretely in Section 6.3 above.

6.5 Consistency Across Languages

MUHC's public presence is bilingual by mandate, and the deployment supports a wider set of languages beyond English and French. To verify that the authorization-first behavior demonstrated in Section 6.3 holds outside English, the identical multi-part surgery-preparation question was submitted to the live deployment in French, Chinese, and Hindi.

Figure 15. The same multi-part surgery-preparation question, submitted natively in French, Chinese, and Hindi. Each response opens with the question as entered by the user and proceeds to a structured, sourced answer in that language, without a separate deployment or translation layer.

The system produced a coherent, correctly sourced, natively written response in each language tested. This matters operationally as much as technically: a governance architecture that only performs well in English would not meet MUHC's actual bilingual mandate, let alone serve the full range of languages spoken by its patient population.

7. Governance, Privacy, and Institutional Control



Figure 16. Governance, privacy, and institutional control: the six pillars of the COMPAiSS institutional framework.

COMPAiSS also makes a strong institutional adoption argument by emphasizing privacy and control. The system uses privacy-by-design principles, does not collect patient names or IDs, operates on Canadian-hosted infrastructure, and enforces scope control through rate limiting and adversarial-input handling. These are the kinds of design choices healthcare organizations need to evaluate early, not after deployment.

A particularly important point is that MUHC retains authority over the greenlist of approved sources. That means the institution is not handing over its knowledge environment to a vendor. It is preserving governance over what the system can say and where the system can look for evidence.

This is a major advantage for public institutions. It allows AI to be introduced without replacing existing authority structures. In practice, that makes the system easier to defend operationally, ethically, and administratively.

8. Greenlist Control and Institutional Authority

The preceding section states that MUHC retains authority over the greenlist. This section shows what that authority actually looks like in practice: who can change it, how quickly, and what happens when something in it breaks.

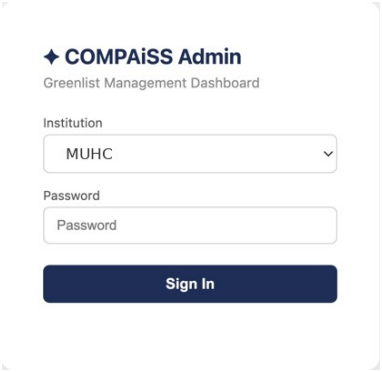


Figure 17. Authenticated access to the MUHC greenlist dashboard. Access is scoped to the institution; only authorized MUHC staff can sign in to view or modify MUHC's approved source list.

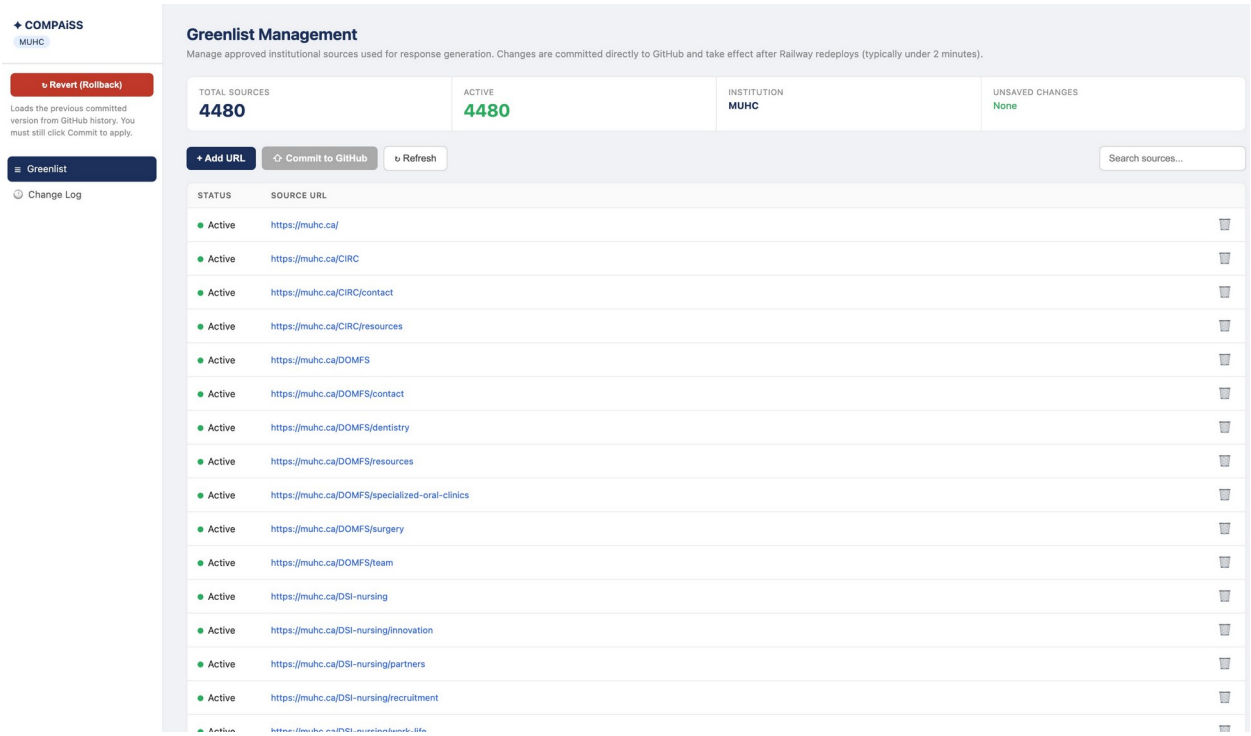


*Figure 18. The greenlist management interface. Every **approved patient-education** facing source is listed individually, with status, and can be added, removed, or reviewed directly by institutional staff without developer involvement. Changes commit directly and take effect on redeploy, typically under two minutes.*

The greenlist is the governance foundation of the deployment. It defines precisely which institutional sources COMPAiSS is authorized to draw from, and nothing else. Authorized staff access a secure, browser-based dashboard to view, add, and manage approved sources directly; no technical expertise is required, changes take effect within minutes of being committed, and the complete version history of every modification is retained for audit.

The full list is also audited on an ongoing basis: checked regularly for broken links, outdated pages, and redirects, with results shared with the institution directly. Source gaps identified during testing or audit are closed within hours rather than routed through an internal IT queue. During a pilot, a single authorized institutional contact typically manages the list in coordination with COMPAiSS; as a deployment matures, that authority transitions to the institution's own designated staff.

Why this is not just a smaller version of the same trust problem

Conventional retrieval systems are built for ingestion convenience: point a crawler at a document estate and index everything reachable. Nobody can then say with confidence which of those documents were actually reviewed and approved for AI use, only that they were technically reachable. A greenlist deployment gives up that ingestion convenience in exchange for provenance control: knowing, for every source the system can cite, exactly who authorized it and when. This also changes what a governance failure looks like when one occurs. In conventional retrieval, an outdated policy or a document nobody meant to include typically surfaces as an incorrect answer delivered to a user, discovered only after the

fact. In a COMPAiSS deployment, the same underlying failure surfaces as a missing answer, flagged for correction during routine validation, before it ever reaches a user.

Scale is a common objection to this model, and it is worth addressing directly: the greenlist does not ask MUHC to build a new governance structure. It mirrors the one MUHC already runs. Patient education content already belongs to Clinical Governance; medication information already belongs to Pharmacy; infection control guidance already belongs to Infection Prevention. None of that ownership is created for COMPAiSS. A greenlist deployment scales the same way MUHC's existing content governance already scales, across sites and departments, by adding the next already-existing content owner to a list, not by re-engineering the system underneath them. A multi-site network does not require multiplying the governance effort; it requires replicating the same dashboard and audit cycle across each site's clinical governance office.

Every addition, removal, and modification is logged with a timestamp and attributed to the person who made it, which is the entire point: governance decisions become attributable rather than invisible. Validation itself remains lightweight regardless of scale, since checking whether an approved source is still reachable is a URL status check, not a document re-read; even at several times MUHC's current greenlist size, a full audit cycle completes in minutes.

9. Deployment Model

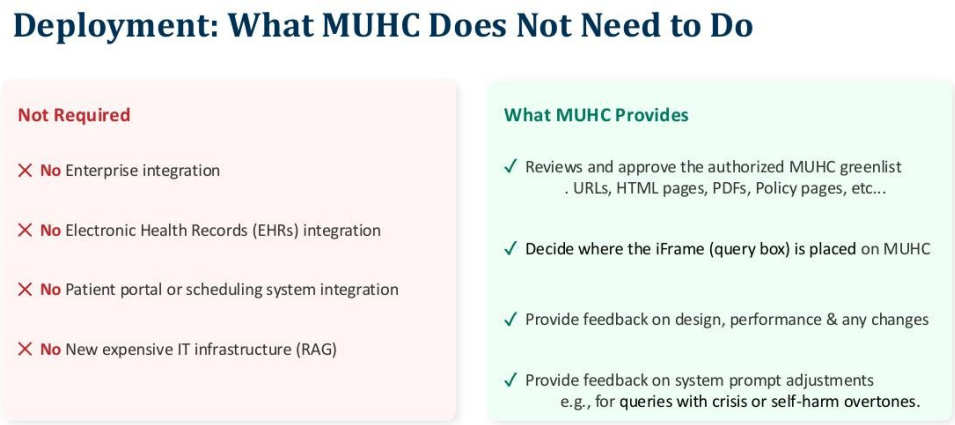


Figure 19. Deployment: What MUHC Does Not Need to Do: no EHR integration, no infrastructure replacement.

Figure 20. A Contained, Low-Risk Way to Test MUHC COMPAiSS: two-week turnaround, no procurement decision required.

The deployment model is another strength of this approach. MUHC does not need to integrate an EHR, patient portal, or new expensive infrastructure. Instead, the institution would review and approve the authorized source list, decide where the query interface is placed, and provide feedback on prompt or design adjustments.

That is important because implementation complexity is one of the biggest barriers to institutional AI adoption. A low-friction pilot model gives the hospital a way to test value without committing to a major systems overhaul: a two-week turnaround pilot using public MUHC content, framed as an evaluation exercise rather than a procurement-heavy transformation.

For the case study, this matters because it shows that authorization-first AI can be introduced incrementally. The institution can test whether the system preserves authority, improves service, and reduces pressure on call centres or static FAQ tools before scaling further.

10. MUHC Operational Scale: Context for the ROI Case

The architectural argument above establishes why authorization-first AI is the correct governance model. This section grounds that argument in MUHC's own publicly documented operational scale, and is deliberately explicit about the line between measured fact and planning assumption, a distinction the ROI figures elsewhere in this deployment package depend on.

10.1 What is publicly documented

MUHC's centralized contact centre is publicly reported to handle approximately 75,000 calls per month, operating 24 hours a day, seven days a week, with a staff of roughly 30. Extrapolated, that is approximately 900,000 calls annually, or roughly 2,465 calls per day. This centralized function sits within

a much larger patient-communication footprint: MUHC reports more than 500,000 outpatient visits annually, hundreds of thousands of emergency department visits, eight major hospital sites, and approximately 14,500 employees.

MUHC has also already invested in centralized appointment services, digital referral systems, SMS appointment reminders, patient portals, and an expanded centralized contact centre specifically designed to resolve inquiries without transferring callers to individual clinics. The publicly documented scope of that centralized service includes appointment booking, rescheduling, cancellations, clinic locations, room confirmation, and referral-portal support, a scope that is predominantly administrative rather than clinical.

What is not publicly available is the operational detail that would be required to model a precise ROI case: average handle time, average speed of answer, abandonment rate, first-call resolution, wait times, the administrative-versus-clinical split of call volume, or cost per call. Those figures would need to come from MUHC directly if a pilot proceeds, and are not asserted here. Although a precise, MUHC-specific ROI case is not yet possible without that data, general ROI estimates based on comparable healthcare deployments are available and can inform planning in the interim.

Separately, for sizing and pricing discussion rather than as an observed statistic, a broader planning assumption is worth stating explicitly and labelling as such: when the full communication ecosystem is considered, telephone, referrals, appointment workflows, and related digital-channel interactions, not the contact centre alone, total annual contacts are more plausibly in the range of 1 to 3 million. This figure should not be presented alongside the measured 900,000-call contact-centre figure as though the two carry the same evidentiary weight. One is documented; the other is a planning estimate.

10.2 Published healthcare benchmarks used as planning context

In the absence of MUHC-specific handle-time data, this case study applies published, cross-referenced healthcare contact-centre benchmarks as illustrative planning context, not as MUHC-specific measurements. Typical healthcare Average Handle Time converges around 6.6 minutes across independent benchmark sources; complex scheduling, referral, or insurance calls often run 8-10 minutes or longer; and general contact-centre benchmarks across all industries place most calls in the 4-10 minute range, with healthcare running structurally above the all-industry average due to verification, documentation, and coordination requirements.

Applied illustratively to MUHC's documented call volume, approximately 900,000 calls per year at 6.6 minutes per call implies roughly 5.94 million minutes, or approximately 99,000 hours, of annual agent conversation time, roughly 48 full-time-equivalent years of agent time devoted to conversation alone. This is explicitly not a staffing estimate: actual staffing also depends on breaks, occupancy, after-call work, scheduling, and 24/7 coverage requirements that this figure does not capture. It illustrates scale, not a claimed savings figure.

10.3 Why handle time is not the right primary metric

The more useful conclusion from this analysis is that average handle time should not be treated as the primary value proposition for authorization-first AI in a hospital contact centre. The relevant question is not whether every call becomes shorter, but which calls consume that time. Parking, directions, appointment location, visitor policies, institutionally authorized clinic preparation instructions, medical records process, and partially automatable appointment changes are better suited to governed AI assistance. Complex scheduling conflicts, emotional or distressed callers, clinical triage, care coordination, and escalations still require human judgement.

That reframing points to a broader observation worth stating plainly: the contact-centre literature overwhelmingly measures speed (handle time, answer time, abandonment) but rarely measures whether the caller received the institutionally correct answer. Operational efficiency metrics and governance metrics are complementary, not interchangeable, and an authorization-first architecture is the piece of that picture current benchmarking practice does not capture.

10.4 The strategic framing this supports

The strongest operational conclusion from this analysis is not a number. MUHC has already centralized and modernized substantial parts of its patient-communication infrastructure: centralized scheduling, digital referrals, SMS reminders, and an expanded contact centre. Within that context, authorization-first conversational AI is not an additional chatbot layered onto existing infrastructure. It is a governed conversational interface to infrastructure MUHC has already built, extending an investment already underway rather than replacing or duplicating it.

11. COMPAiSS Fills the Gap: Between FAQ Bots and General AI

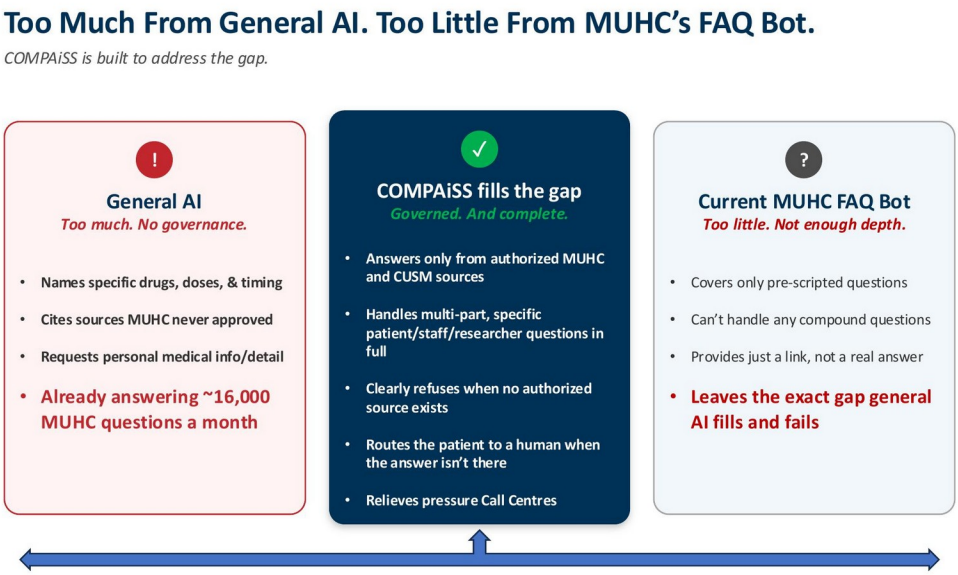


Figure 21. Too much from general AI. Too little from MUHC's FAQ bot. COMPAiSS positioned as the governed answer between the two, resolving the dilemma introduced in Section 6.4.

Note: the approximately 16,000 figure shown in the graphic refers to the estimated reduction in monthly visits to muhc.ca between April and May 2026. It is not an estimate of MUHC-related questions answered by AI, unique patients, or clinical errors.

The "too much / too little" framing is one of the clearest ways to explain the market gap, and it makes a directional claim, not just a feature comparison: general AI is too permissive, answering with details it should not be giving, while the current scripted FAQ bot is too narrow, covering only pre-written questions and failing on any real, multi-part query.

That direction of movement is not asserted in isolation. It is consistent with Section 3's traffic evidence and Figure 3's independent research (68% of Google searches now ending without a click; 32% of U.S. adults already turning to AI chatbots for health information) and with the estimated 16,000 fewer monthly

visits to muhc.ca. The estimated reduction in visits is consistent with the possibility of channel substitution documented elsewhere in healthcare. It does not establish that the lost visits became AI interactions. A FAQ bot that cannot handle a real question does not suppress demand for AI-mediated answers. It routes that demand to a system the institution never authorized.

COMPAiSS is positioned as the governance response to that specific movement, not merely a better version of either tool. Users want more than links, but the institution cannot allow open-ended generation that drifts beyond approved sources. The case study therefore shows that the problem is not "AI versus no AI." It is "governed AI versus ungoverned AI," and the FAQ bot's limitations are what make that choice active rather than hypothetical.

12. Discussion

The MUHC case offers a broader lesson for healthcare institutions. The right question is not whether AI can answer questions quickly. The right question is whether the institution can preserve authority over the answer while still delivering a useful experience.

COMPAiSS represents an architectural response to that challenge. By placing authorization before substantive answer generation, it converts institutional policy from a post-hoc filter into a structural requirement. That is especially relevant for hospitals, where errors are not merely inconvenient. They can influence clinical preparation, patient behavior, and institutional trust.

The live deployment evidence in Section 6 also illustrates a second, related benefit that is easy to overstate but worth naming carefully: an authorization-first system does not just avoid unauthorized answers, it makes its own limits visible and traceable. Where the system's answer was thinner than a general AI's would have been, that thinness was explainable by reference to a specific, named source gap, not hidden behind a fluent-sounding non-answer. That auditability is itself a governance property, distinct from and additional to the accuracy of any single answer.

13. Conclusion

The MUHC deployment case shows why authorization-first AI is well suited to regulated healthcare settings. MUHC has a complex digital footprint, broad user demand, and a growing challenge from general-purpose AI tools that answer outside institutional control. It has also already invested substantially in centralizing and modernizing its own patient-communication infrastructure. In that environment, a system that checks authorization before generation, and that extends existing infrastructure rather than replacing it, is a meaningful governance improvement.

COMPAiSS is not presented as a better chatbot. It is presented as a different institutional architecture. That difference matters because it allows hospitals to give users useful answers while maintaining control over source authority, privacy, and accountability.

Core architectural distinction

Authorization-first AI checks whether a query can be answered from approved institutional sources before generation is permitted to run. Conventional RAG retrieves material before generation, but retrieval alone does not establish an externally enforced decision that sufficient institution-authorized evidence exists before substantive answering is admitted. The proposed distinction is the authority-and-evidence condition controlling substantive answer generation, not a claim that retrieval normally follows generation or that no other system has upstream controls. For regulated healthcare information, that sequence is the

difference between a defensible answer and an institutional liability.

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